

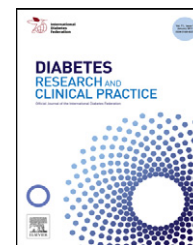


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Anti-diabetic effect of standardized herbal formula PM021 consisting of *Mori Folium* and *Aurantii Fructus* on type II diabetic Otsuka Long–Evans Tokushima Fatty (OLETF) rats

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ABSTRACT

PM021, which consists of two herbal components, *Mori Folium* and *Aurantii Fructus*, is routinely used to treat diabetes in Korea. In this study, the anti-diabetic effect of PM021 on an animal model of developing type 2 diabetes of Otsuka Long–Evans Tokushima Fatty (OLETF) rats was investigated. Eight weeks of age male OLETF rats were treated daily with PM021 or vehicle for five months. Specifically, changes in body weight, blood glucose, urine volume, food intake and oral glucose tolerance were measured in rats for five months. The rats in this study were divided into four groups: a Long–Evans Tokushima Otsuka (LETO) rat group, which is a genetic control group for OLETF, that received no treatment; a PM021 treatment group of LETO rats; OLETF rats that received no treatment; and OLETF rats that received PM021 treatment. The results showed that PM021 significantly prevented increases in body weight, blood glucose, and urine and food intake that resulted from the induction of obesity and diabetes. PM021 also improved glucose tolerance in OLETO rats. However, PM021 had no effect on LETO rats, a control group of OLETF rats. Taken together, these findings indicate that PM021 has distinct anti-diabetic effects without any adverse effects or toxicities.

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1. Introduction

Due to the adverse effects associated with conventional anti-diabetic drugs [1,2], many trials have recently been conducted to identify and develop new anti-diabetic drugs from herbal medicines to minimize side effects. Numerous animal and

clinical studies of various herbal medicines have been conducted, and some have reported significant improvements in controlling diabetic symptoms without any noticeable adverse effects [3]. PM021 is composed of *Mori Folium* and *Aurantii Fructus*. *Mori Folium* (MF), which is the leaves of *Morus alba* L. (Moraceae), is one of the most widely consumed medicinal herbs in Asia, and its medications have been found

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to be useful in the treatment of diabetic hyperglycemia [4]. Different parts of the mulberry tree have also been applied in the clinical treatment of various diseases. Recent evidence shows that the leaves and roots of the mulberry tree possess several medicinal properties including hypoglycemic, hypotensive and diuretic effects [5]. In addition, it has been shown that MF is clinically effective for the treatment and prevention of diabetes [6]. *Aurantii Fructus* (*Citrus aurantium* L. Rutaceae) (AF) has frequently been used to treat cardiovascular symptoms in Asian countries. It has been reported that AF reduces portal pressure in portal hypertensive rats through increased mean arterial pressure, that it prevents infective shock via increased cardiac output and that it inhibits blood coagulation via an anti-fibrinolytic effect [7]. Other potential therapeutic effects of AF extracts, such as hypocholesterolemic actions through HMG-CoA reductase inhibition [8], hypolipidemic actions [9] and anti-asthma action via the inhibition of Ig E production [10], have also been reported. It is also well known that several flavonoids are included in AF, including naringin, which is known to have anti-ischemic action [11], and synephrine, which is known to possess anti-shock activity [12]. Naringin and hesperidin are also known for their anti-diabetic [13,14] antioxidant [15] and cholesterol-lowering activities [9]. Other potential therapeutic effects of hesperidin, such as relaxation, have also been reported [16]. These functions of AF suggest that it can serve as an effective anti-diabetic agent to improve hyperglycemia and recover dysfunction of the cardiovascular system induced by diabetes.

Type-II diabetes is a disease characterized by chronic hyperglycemia resulting from abnormalities in glucose metabolism and insulin resistance [17]. OLETF rats, which have been established as an animal model of human type II diabetes, exhibit obesity, hyperinsulinemia, hypertriglyceridemia, and hyperglycemia [18]. The brain–gut peptide cholecystokinin (CCK) inhibits food intake following peripheral or site directed central administration. Peripheral exogenous CCK inhibits food intake by reducing the size and duration of a meal. Antagonist studies have demonstrated that the actions of the exogenous peptide mimic those of endogenous CCK. Antagonist administration results in increased meal size and meal duration [19]. The inhibitory actions of CCK on feeding are mediated through interactions with CCK-1 receptors. The recent identification of the OLETF rat as a spontaneous CCK-1 receptor knockout model has allowed a more comprehensive evaluation of the feeding actions of CCK. OLETF rats become obese and develop type 2 diabetes [20]. A spontaneously diabetic rat with polyuria, polydipsia and obesity was discovered at the Tokushima Research Institute in 1984 in an out-bred colony of Long-Evans rats that had been purchased from Charles River a few years earlier. Through selective breeding, a line of rats characterized by obesity, adult onset hyperglycemia, chronic diabetes mellitus and late conversion from NIDDM to insulin dependent diabetes mellitus (IDDM) was established by 1991 [18]. These OLETF rats have been extensively studied as a model of human NIDDM [21]. Both male and female OLETF rats develop obesity. Starting from normal weights at the time of weaning, male and female OLETF rats become 30–40% heavier than LETO controls by 20 weeks of age [22]. Plasma glucose levels following oral glucose loads are elevated as early as eight

weeks of age and by 24 weeks of age the rats are hyperglycemic and hyperinsulinemic. Many of the males develop IDDM by 65 weeks of age [20].

Based on these observations, the anti-diabetic effect of PM021 on an animal model of developing type 2 diabetes of Otsuka Long-Evans Tokushima Fatty (OLETF) rats was investigated through analyses of changes in body weight, blood glucose, urine volume, and food intake of rats, as well as an oral glucose tolerance test (OGTT) in rats.

2. Materials and methods

2.1. Preparation of PM021

All of the sprayed dried extracts of PM021 components used in this study were purchased from Sun-Ten Pharmaceutical Company (Taipei, Taiwan). One gram of AF powder contained 0.7 g of AF extract and 0.3 g of a raw herb powder, while 1 g of MF contained 0.7 g of MF and 0.3 g of a raw herb powder. AF and MF powders were mixed equally based on quantity.

2.2. Quantitative chromatographic analysis

HPLC analysis was conducted using the Waters system (Waters Co., Milford, MA, USA) with a 717+ auto-sampler, 2996 photodiode array detector (PDA) 2487 dual λ absorbance detector and a 1525 binary HPLC pump. A Waters Millennium System (Waters Co., Milford, MA, USA) was used for data acquisition and integration. HPLC grade and other reagents (J.T. Baker Co., Ltd., Phillipsburg, NJ, USA) were used for HPLC analysis. All solvents were filtered and degassed prior to use. The MF and AF mixture (PM021) were accurately weighed to 1 g and then dissolved in 40 ml of methanol. The sample was then subjected to ultrasonic treatment for 60 min and centrifuged at 15,000 rpm for 10 min, after which the supernatant of the sample was filtered through a 0.45 μ m syringe filter (PVDF, GELMAN, USA). For quantitative analysis of PM021, astragaloside and naringin were purchased from Chroma-Dex for use as standard materials for *Mori Folium* and *Aurantii Fructus*, respectively.

The separation was conducted on a reverse phase system (XTerra™ C₁₈, 4.6 mm $\times$ 150 mm i.d., 5 μ m, Waters Co., USA). A gradient mobile phase consisting of A (0.1% formic acid in acetonitrile) and B (water) was used to run the separation using the following elution program: from 100% B to 100% A in 30 min, gradient system; flow rate, 1.0 ml/min; injection volume, 25 μ l. The samples were detected using a Photodiode Array with detection wavelengths of 260 nm and 283 nm. The calibration curves of astragaloside and naringin were linear in the range of 50–313 μ g and 500–313 μ g, respectively.

2.3. Animals and treatment

Age-matched male Otsuka Long-Evans Tokushima Fatty (OLETF) rats, a model of spontaneous non-insulin dependent diabetes mellitus, and male Long-Evans Tokushima Otsuka (LETO) rats, the non-diabetic control model of OLETF rats, were kindly provided by Otsuka Pharmaceutical Co., Tokushima, Japan. The rats were individually housed and allowed free

access to feed and tap water under strictly controlled and pathogen free conditions (room temperature: $23 \pm 1^\circ\text{C}$, relative humidity: $50 \pm 10\%$, light cycle: 07:00–19:00). The rats were fed a standard rodent pellet chow and acclimatized to their environment for two weeks prior to commencement of the experiments.

Forty two rats that were eight weeks of age were randomly assigned into four subgroups and then treated daily with PM021 or vehicle for five months: (1) LETO controls ($n = 10$) received vehicle (saline), (2) LETO PM021-treated rats ($n = 11$) received PM021 (500 mg/kg in saline), (3) OLETF controls ($n = 10$) received vehicle (saline), (4) OLETF PM021-treated rats ($n = 11$) received PM021 (500 mg/kg in saline). The experimental procedures were conducted according to the animal care guidelines of the NIH and the Kyung Hee University Institutional Animal Care.

2.4. Functional protocol

The body weight of each rat was recorded at the beginning of the experiment and then at 3 months and 5 months. Food intake was determined by measuring the difference between the pre-weighed chows and the weight of the food that remained every 24 h during the last 3 and 5 months of the 5-month-diet period. Blood samples were collected from the tail vein at 0 months, 3 months and 5 months of treatment, and the resulting serum was stored at -20°C until analysis. The concentration of plasma glucose was assessed using a Gluco-Tester (Life Scan Inc., USA) to analyze the serum of each treatment group. An oral glucose tolerance test (OGTT) was conducted on rats that had fasted for 16 h at 0, 3 and 5-months. Glucose (1 g/kg body weight) was administered to rats orally, and blood samples were collected at 30, 60, 90 and 120 min after the OGTT from the tail vein as previously described [23]. All animals were caged individually for 24-h urine collection at 0, 3 and 5-months.

2.5. A 4-week oral toxicity dose-range finding study of PM021 in SD rats

This study was carried out to investigate the rough toxicity of PM021 in Sprague–Dawley (SD) rats when administered orally by gavage for 4 weeks repeatedly. After PM021 was administered orally at dose levels of 500, 1000 and 2000 mg/kg/day to 10 per group (male 5, female 5) for 4 weeks, several parameters such as mortality, clinical signs, body weight changes, food & water consumption, urinalysis, hematology & serum biochemistry examinations, necropsy findings and absolute & relative organ weights were observed. This study was conducted in accordance with the two standards: The Korea Food and Drug Administration (KFDA) Notification No. 2005-60 'The Standards of Toxicity Study for Medicinal Products' (October 21, 2005), and The KFDA Notification No. 2005-79 'Good Laboratory Practice (GLP)' (December 21, 2005).

2.6. Statistical analysis

The results are presented as the mean $\pm$ S.E.M. The effect of PM021 on each parameter was examined by one-way analysis of variance (ANOVA). Individual differences among groups

were analyzed by Dunnett's test and those at $p < 0.05$ were considered to be significant.

3. Results

3.1. Quantitative analysis of PM021

For quality assurance, three different batches of PM021 were analyzed using astragalin and naringin as standard materials. The relationship between the concentration and the peak-

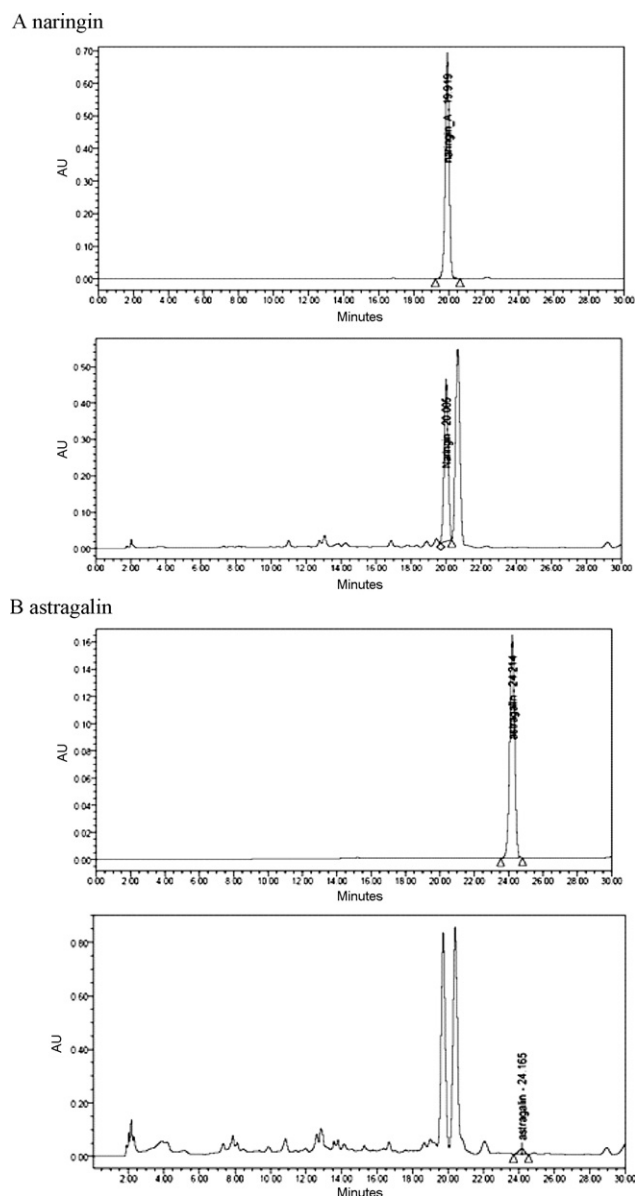


Fig. 1 – Quantitative analysis of PM021. (A) HPLC chromatogram of standard naringin, which is a major compound of Aurantii Fructus at 283 nm (top). HPLC chromatogram of PM021 for determination of naringin (RT = about 20.0) (below). (B) HPLC chromatogram of standard astragalin, which is a major compound of Mori Folium, at 260 nm (top). HPLC chromatogram of PM021 for determination of astragalin (RT = about 24.1) (below).

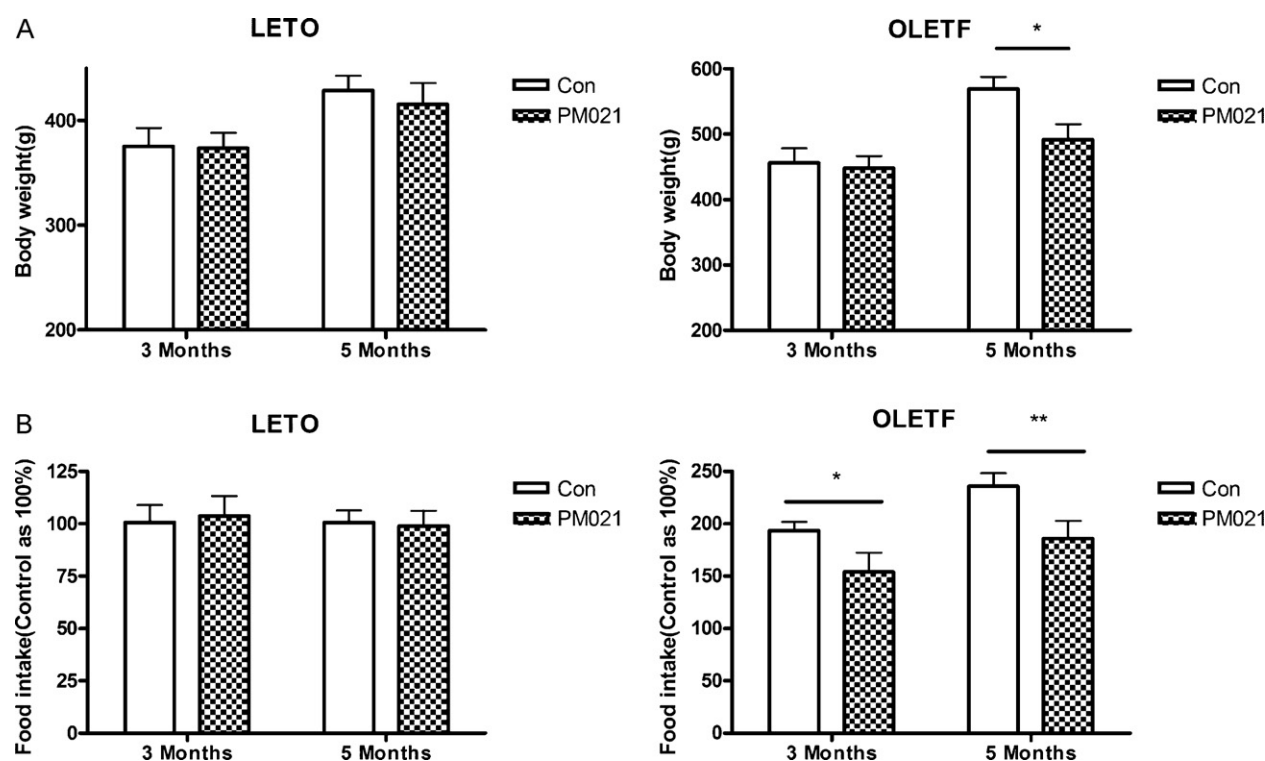


Fig. 2 – Effect of PM021 on body weight (A) and food intake (B) of OLETF and LETO rats at 3 or 5 months. Data were obtained from 10 or 11 animals per group. 3 months of control LETO rat as 100%. Significantly different from OLETF control rats (* $p < 0.01$, ** $p < 0.001$).

area was measured by the minimum square method (R^2 value). The average content of astragalin and naringin in PM021 was $265.99 \pm 4.842 \mu\text{g/g}$ ($n = 3$) and $35.409 \pm 0.439 \text{ mg/g}$ ($n = 3$), respectively, based on the above formula (Fig. 1).

3.2. Effect of PM021 on body weight and food intake in OLETF rats

As shown in Fig. 2, the body weights of OLETF rats increased at higher speed compared with LETO control rats, although the body weights of LETO rats also increased gradually with age. PM021 treatment significantly reduced the body weights in OLETF rats at 5 months (569.2 ± 18.6 vs. 491.7 ± 23.5 , $p < 0.01$). Conversely, PM021 treatment significantly decreased the food intake (193.5 ± 8.3 vs. 153.7 ± 18.5 at 3 months; 235.7 ± 12.8 vs. 185.8 ± 16.8 at 5 months, $p < 0.01$, see Fig. 2). These findings indicate that PM021 may improve obesity due to OLETF.

3.3. Effect of PM021 on blood glucose and urine output of OLETF and LETO rats

Fig. 3 shows the effects of PM021 on blood glucose. When compared with LETO rats, OLETF rats had significantly increased blood glucose levels at 3 and 5 months. However, PM021 treatment led to significantly reduced blood glucose levels in OLETF rats at 5 months (125.6 ± 5.2 vs. 109.3 ± 6.8 , $p < 0.01$, see Fig. 3). We also examined urine output, which is one of the major characteristics that can be used to identify diabetes. Urine output was two or three folds greater in OLETF

rats at 3 or 5 months, respectively, when compared with that of OLETF rats. PM021 treatment decreased the urine output at 3 and 5 months (182.5 ± 8.1 vs. 151.7 ± 9.6 for 3 months; 288.3 ± 14.2 vs. 228.6 ± 15.3 for 5 months, $p < 0.01$, see Fig. 3).

3.4. Effect of PM021 on oral glucose tolerance test (OGTT) of OLETF rats

Rats underwent an oral-glucose-tolerance test (OGTT) after an overnight fast. PM021-treated OLETF rats also showed significantly decreased blood glucose levels, as indicated by the plasma glucose response in the OGTT, when compared to vehicle-treated OLETF rats in both the 3 and 5 month treatment groups (Fig. 4). These results suggest that PM021 could improve poor plasma glucose responses and eventually decrease the blood glucose levels attributed to OLETF.

3.5. 4 weeks oral toxicity dose-range finding study of PM021 in SD rats

Repeated oral administration of PM210 to rats for 4 weeks resulted in no toxicological changes in any of examinations, namely mortality, clinical signs, body weight changes, food & water consumption, urinalysis, hematology & serum biochemistry examinations, necropsy findings and absolute & relative organ weights, within the range of 500–2000 mg/kg/day doses. Thus, the no observed adverse effect level (NOAEL) of PM021 is considered to be 2000 mg/kg/day for both males and females (supplementary data 1).

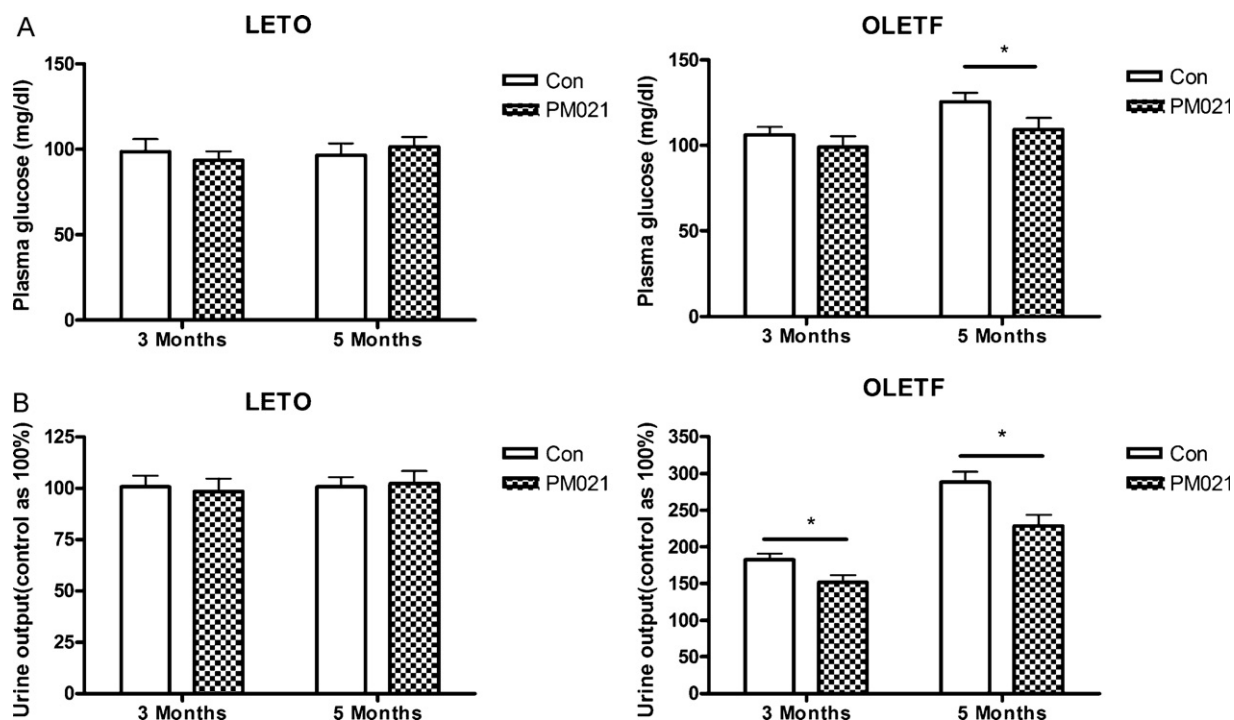


Fig. 3 – Effect of PM021 on plasma glucose (A) and urine output (B) of OLETF and LETO rats for 3 or 5 months. Data were obtained from 10 or 11 animals per group. 3 months of control LETO rat as 100%. *Significantly different from OLETF control rats ($p < 0.01$).

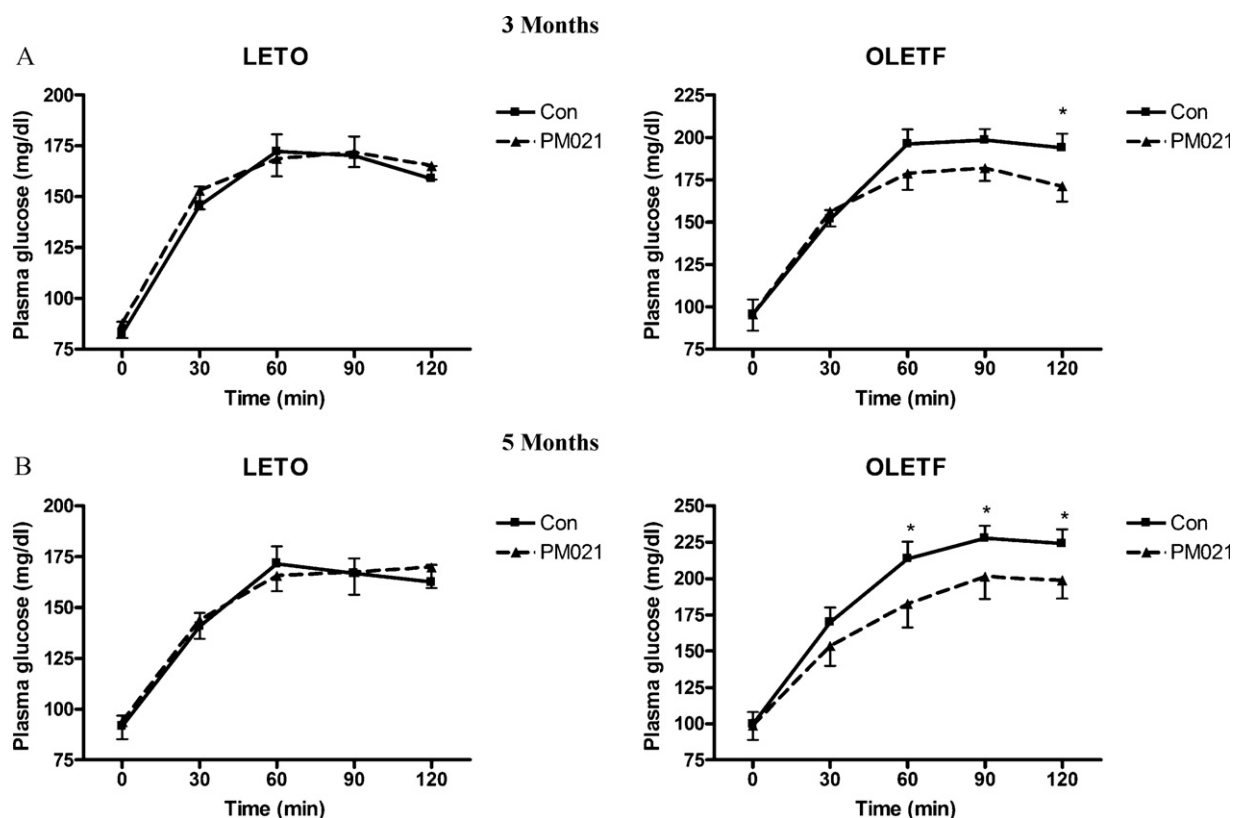


Fig. 4 – Effect of PM021 on plasma glucose concentrations during an oral glucose tolerance test (OGTT) in OLETF and LETO rats treated with PM021 or vehicle for 3 (A) or 5 months (B). *Significantly different from OLETF control rats ($p < 0.01$).

4. Discussion

In the present study, the blood glucose and food intake levels increased with the passage of time in OLETF rats fed the control diet, and symptoms of diabetes such as increased food intake, hyperuria, obesity, and hyperglycemia appeared in the 3rd month after commencing feeding with the experimental diets. These symptoms in the OLETF rats were significantly improved after PM021 treatment. It was previously reported that hesperidin and naringin, major components of AF, were found to be very effective at improving the lipid metabolism of a type II diabetic animal model by altering the hepatic lipid metabolizing enzyme activities and fecal lipids [13]. In addition, the compounds were apparently beneficial for lowering the blood glucose level via up-regulation of the hepatic glucokinase and peroxisome proliferator-activated receptor γ and adipocyte glucose transporter 4 [14]. As such, these coordinated responses of hesperidin or naringin supplement likely play an important role in regulating the lipid and glucose metabolism in type II diabetic animal models. Accordingly, both compounds exhibited antidiabetic properties and were able to reduce certain diabetic complications related to hyperlipidemia [14]. Moreover, 1-deoxynojirimycin, a major component of MF, is well known for its hypoglycemic effects, which occur through α -glucosidase inhibitory activity and lead to inhibition of the absorption of glucose in the intestine [4].

In addition to regulate the lipid and glucose metabolism by its components, PM021 reduced food intake and resulted in inhibition of body weight gain in obese OLETF rats. According to our results, body weight gain and food intake reduction by PM021 were effective only in obese OLETF rats. SD rats and LETO rats were not shown any significant changes on body weight gain and food intake by PM021 (Fig. 2 and supplementary data 1). Although we suggest that these effects would be similar to those of sibutramine (ReductilTM), an appetite-lowering, anti-obesity drug, it remains to be elucidated the precise mechanisms of PM021 to further study. Previously, we compared individual extracts and PM021 in streptozotocin induced diabetic rats. Although AF and MF were effective on blood glucose decrease and inhibition of body weight loss, PM021 had synergic effects on blood glucose decrease and inhibition of body weight loss (unpublished data). According to the blood glucose lowering effect in OLETF rats and streptozotocin induced diabetic rats by PM021 treatment, we assumed that it would affect on insulin sensitivity or insulin level.

Because AF and MF were allowed to use as an additive in functional food, we predicted that AF and MF would be very low toxicity. As shown in supplementary data 1, PM021 had no significant toxicity up to 2 g/kg in a 4 weeks oral DRF study.

In conclusion, the results of this study revealed that PM021 induced a hypoglycemic effect via improvement of glucose tolerance. These findings indicate that there is the potential for the development of a novel functional food containing PM021 for suppression hyperglycemia. Although we focused on the diabetic symptoms such as body weight change, blood glucose, urine volume, food intake and OGTT by PM021 treatment, it also needs to analyze another diabetic diagnosis

factors such as glycated hemoglobins (HbA1c) and the insulin response to clarify its effects. In the normal 120-day lifespan of the erythrocytes, glucose molecules react with hemoglobin, forming glycated hemoglobin. Once a hemoglobin molecule is glycated, it remains that way. HbA1c represents the integrated glucose concentration over the preceding 8–12 weeks. It is, therefore, free of the large fluctuations that occur daily in blood glucose concentrations [24]. The OGTT has often been used to evaluate apparent insulin release and insulin resistance in various clinical settings. Because plasma glucose and insulin responses during the OGTT reflect the ability of pancreatic β -cells to secrete insulin and the sensitivity of tissues to insulin [25], the OGTT has also been often used to evaluate β -cell function and insulin resistance. Further studies are being conducted to isolate the active components in PM021 and to elucidate the mechanism of action for its hypoglycemic effect.

Acknowledgment

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Appendix A Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.diabres.2011.03.037](https://doi.org/10.1016/j.diabres.2011.03.037).

Conflict of interest

There are no conflicts of interest.

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